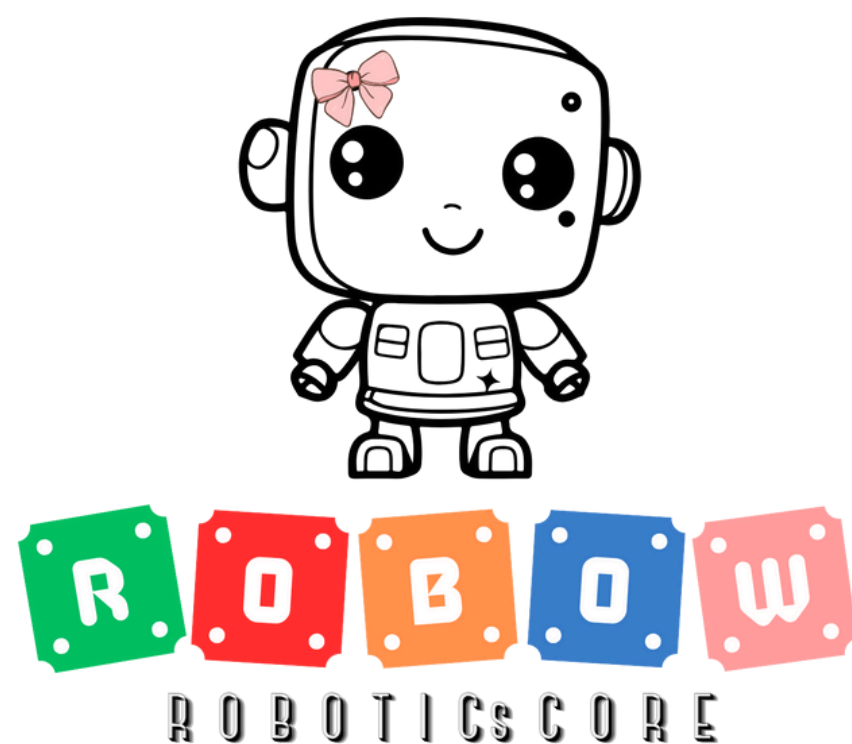




SUTD Invested Venture



Robotics Outreach Body for Optimizing Wonders



SMORPHI²
TRANSFOBOT
RECONFIGURABLE ROBOT

Sri Sri Center for Robotics & Automation
Faculty of Engineering & Technology



~ AGENDA ~

Robotics & Importance
An Introductory Video(s)
Hardware Specifications
Mechanism of Robots
How it Works?
H/W + S/W Integration
Programming
Application

/ >



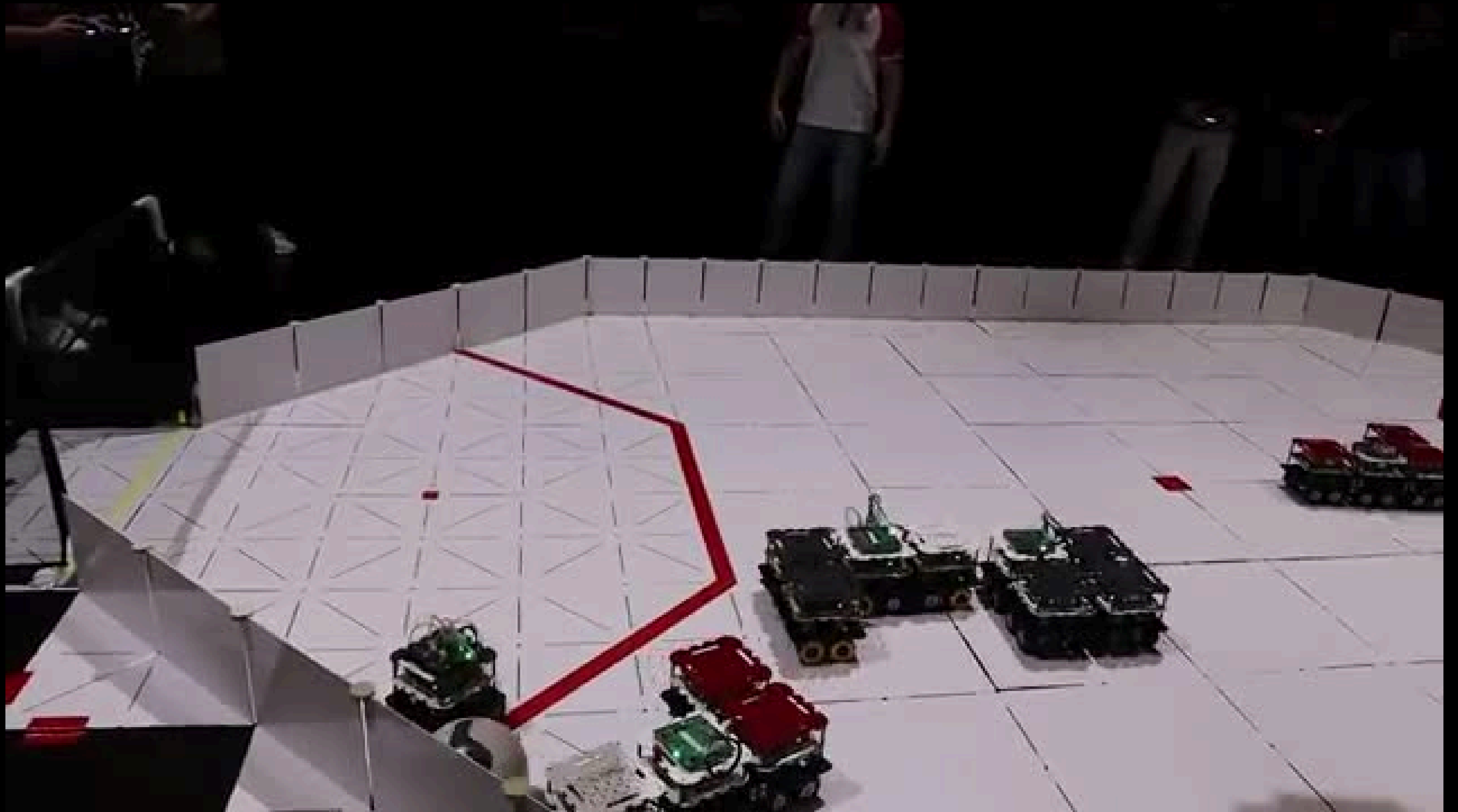
~ INTRODUCTION ~

Reconfigurable robots are *robotic* systems designed with *modular* or *adaptable* structures that allow them to change their **shape, configuration, or functionality** based on the *task at hand*.

~ VIDEO DEMONSTRATION ~

WIT

~ VIDEO DEMONSTRATION ~



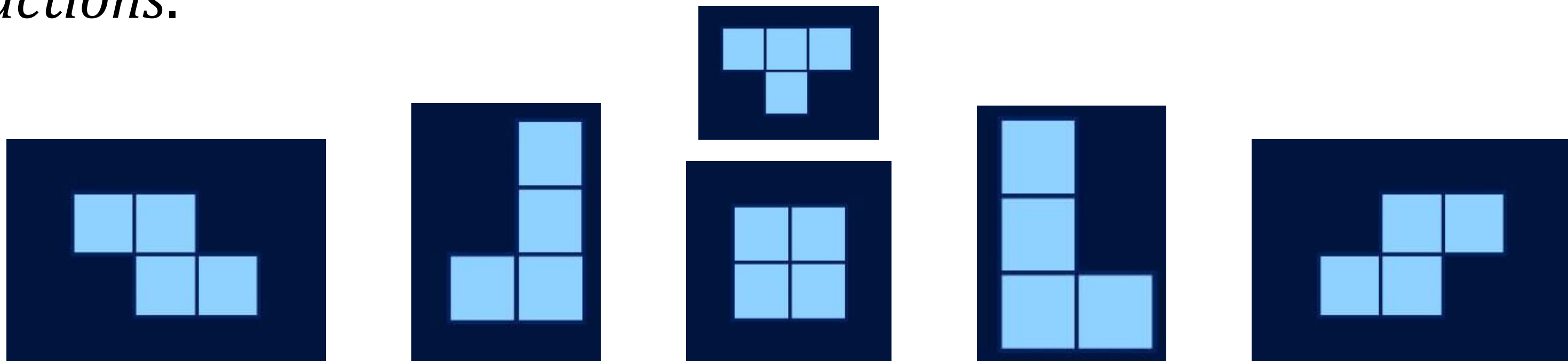
~ HARDWARE SPECIFICATIONS ~

- ***Modular Design Concept***
- Components List: *Multiple* Sensors, Actuators, *Locomotive* Motors
- **Adaptive Chassis and Configurable Joints!**
- **Mecanum Wheels** for Omnidirectional Movement



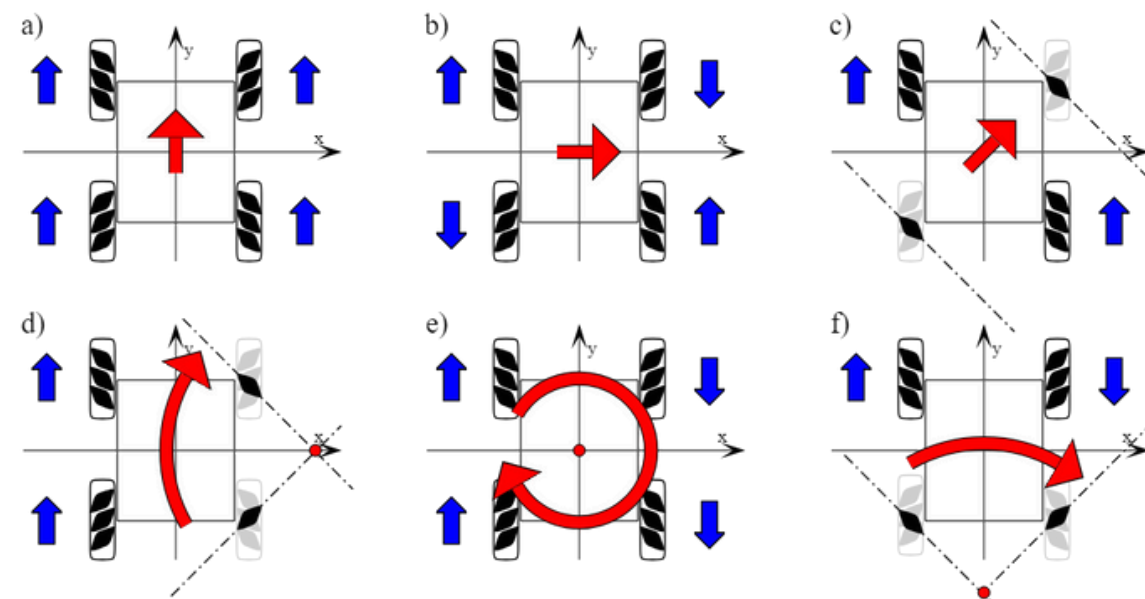
~ MECHANISM ~

- Developed a robot capable of transforming into various shapes, including **O, I, L, T, J, S, and Z**.
- Integrated **Mecanum wheels** for *omnidirectional movement* and a *lock-latch mechanism* for ***seamless shape transitions***.
- Added advanced features like a **sound sensor** and a **camera module**, enabling the robot to *detect sound* and *follow human actions*.



~ HOW THE ROBOT WORKS ~

- Reconfiguration Mechanism: 4 *Adaptive Modules*
- **Real-time Processing and Decision Making**
- **Controlled Algorithms and Motion Planning**
- *Remote Control + Autonomous Operation*





~ HARDWARE & SOFTWARE ~

- **Hardware:** ESP32 Chipset, Li-Po Battery, Wi-Fi/Bluetooth, Numerous Sensors.
- **Software:** *MicroController IDE w/ Compatible Compiler & Git.*
- **Integration:** Seamless hardware-software interaction through Arduino IDE Interface!

~ PROGRAMMING ~

Multiple Program sessions & Longing hours were dedicated to refine the codes for the intensive, and accurate integration of the robot hardware and the BLE Software.

```
#include <smorphi.h>
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>

#define SERVICE_UUID "6E400001-BE12-4F32-B0A0-506000000000"
#define CHARACTERISTIC_UUID_TX "6E400002-BE12-4F32-B0A0-506000000000"
#define CHARACTERISTIC_UUID_RX "6E400003-BE12-4F32-B0A0-506000000000"

BLECharacteristic *pCharacteristic;
bool deviceConnected = false;
Smorphi my_robot;

class MyServerCallbacks : public BLEServerCallbacks {
    void onConnect(BLEServer *pServer) { deviceConnected = true; }
    void onDisconnect(BLEServer *pServer) { deviceConnected = false; }
};

class MyCallbacks : public BLECharacteristicCallbacks {
    void onWrite(BLECharacteristic *pCharacteristic) {
        String data = pCharacteristic->getValue().c_str();
        if (!data.isEmpty()) {
            Serial.println("Received: " + data);
            handleBLEData(data);
        }
    }
};

void handleBLEData(String cmd) {
    if (cmd == "R") my_robot.I();
    else if (cmd == "S") my_robot.O();
    else if (cmd == "T") my_robot.L();
    else if (cmd == "U") my_robot.J();
    else if (cmd == "V") my_robot.Z();
    else if (cmd == "W") my_robot.S();
    else if (cmd == "X") my_robot.T();
}
```

Service + Characteristics ID: Confidential

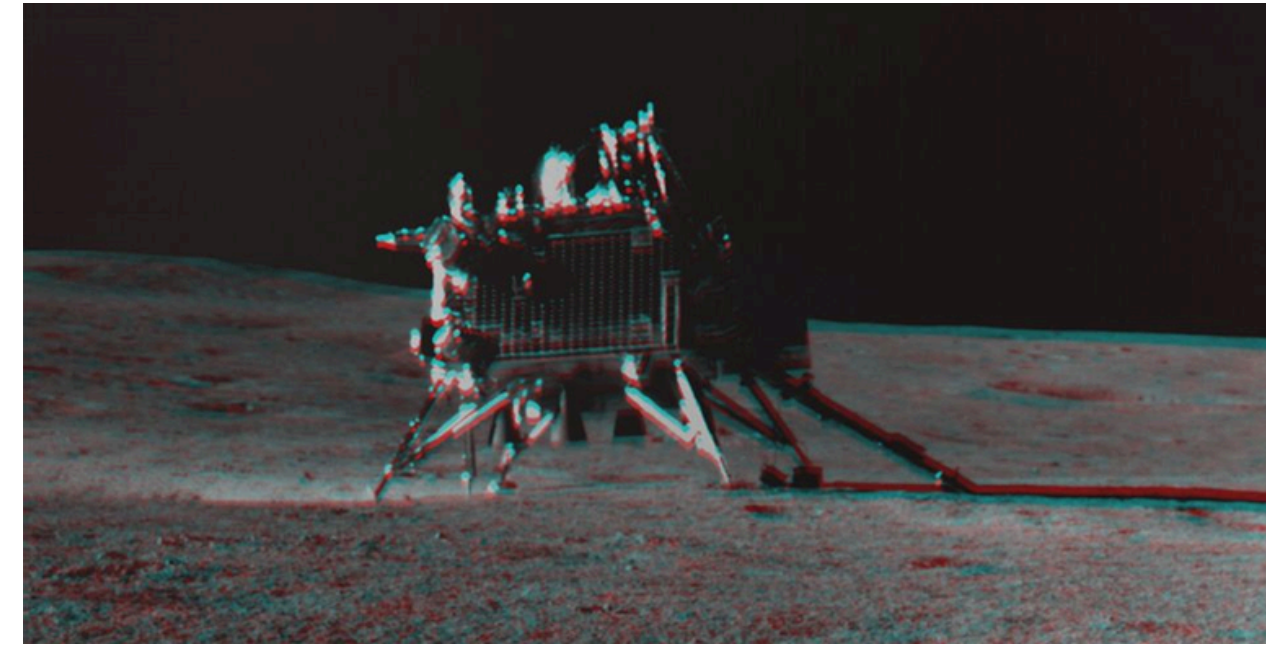
Code Setup:

```
void setup() {
    Serial.begin(115200);
    my_robot.BeginSmorphi();
    BLEDevice::init("Smorphi Square BLE");
    BLEServer *pServer = BLEDevice::createServer();
    pServer->setCallbacks(new MyServerCallbacks());
    BLEService *pService = pServer->createService(SERVICE_UUID);
    pCharacteristic = pService->createCharacteristic(CHARACTERISTIC_UUID_TX, BLECharacteristic::PROPERTY_NOTIFY);
    pCharacteristic->addDescriptor(new BLE2902());
    BLECharacteristic *pRXCharacteristic = pService->createCharacteristic(CHARACTERISTIC_UUID_RX, BLECharacteristic::PROPERTY_WRITE);
    pRXCharacteristic->setCallbacks(new MyCallbacks());
    pService->start();
    pServer->getAdvertising()->start();
    Serial.println("Waiting for client...");
}

void loop() {}
```

~ APPLICATIONS OF ROBOTICS ~

- Mecanum Wheels *inspired* by **Moon Landing Rovers**
- Industrial Applications: **Warehouse Automation**
- Disaster Management: *Search & Rescue* Missions
- **Space Exploration & Adaptive Robotics**





~ EXPERIENCE YOURSELF IN SIMULATION ~

- **SMORPHI IMAGINARY:** *Interactive Reconfigurable Robot Simulation*
- **Hands-On Experience** with *Virtual Prototyping*
- Understand *Motion Dynamics & Control Strategies*



~ JOIN THE MOVEMENT ~

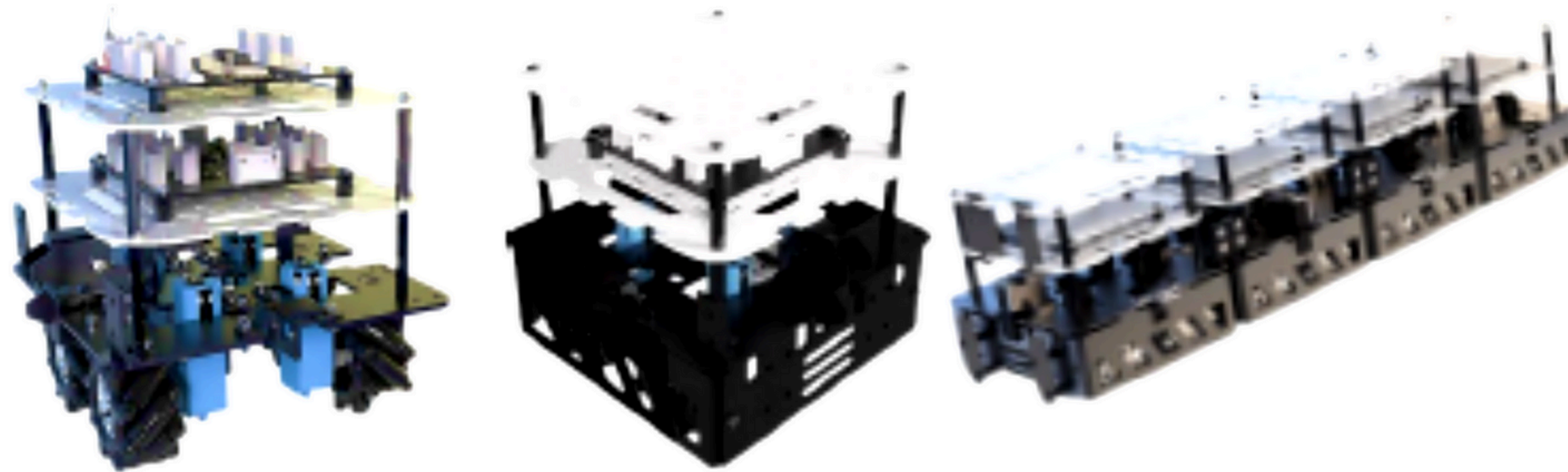
- **Smorphi Imaginary** + Hackathon Opportunities!
- Summer Workshop & Internship!
- Collaboration & Research in *Reconfigurable Robotics*.

[illegible]



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Join Us! Be a part of the next-gen robotic revolution!



In collaboration with
Wefaa Robotics
(SUTD invested venture)

